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Hiring Manager

Integrated Biosciences, Inc.

Redwood City, CA

Dear Hiring Manager,

I'm Ajay Venkatesh, completing an M.S. in Computer Engineering at NYU, with production software experience at PwC in Bengaluru, a summer SDE internship at Amazon, and ongoing on-campus work at NYU's Novel AI Technologies. I'm applying for the Full-stack Software Engineer role, where I'd partner with scientists and engineers to build tools that accelerate small-molecule drug discovery and age-related therapeutics research.

At NYU, I architected full-stack systems from **PostgreSQL** database infrastructure through **Python/Node.js** backends to **React** and **TypeScript** frontends for a life-sciences research platform. I built **ETL pipelines**, researcher-facing dashboards, and **LLM-powered document workflows**, collaborating closely with cross-functional partners to translate experimental needs into scalable software deployed with **Docker** on **AWS**.

At Amazon, I built an **AI agent harness** with **RAG**, an **agentic workflow**, and a **Model Context Protocol (MCP)** server for tool-augmented retrieval, alongside **AWS CDK** infrastructure on **ECS Fargate**. I also shipped a **React/TypeScript** client integrated with backend **REST APIs** and disciplined test coverage. At PwC, I automated structured extraction from scientific documents with an **NLP OCR pipeline** on **Azure**, supporting healthcare analytics workflows used by research-adjacent stakeholders.

My stack includes **TypeScript**, **React**, **Python**, **PostgreSQL**, **Docker**, **AWS**, **Vercel**, **LangChain**, **MCP**, **Cursor**, **GitHub Copilot**, and **Claude Code**. **Computational chemistry**, **wet-lab robotics**, and dedicated **Codex** production workflows are areas I have not owned end-to-end yet, but my life-sciences ML publications, scientist-facing tooling experience, and AI agent infrastructure work give me a strong foundation to ramp on Integrated Biosciences' discovery platform.

What pulls me toward Integrated Biosciences is the chance to build software where engineering quality directly accelerates therapeutic discovery for age-related diseases. I care about great UX for researchers, reliable data infrastructure, and using AI thoughtfully to help scientists move faster without sacrificing rigor.

I'd welcome the chance to discuss further. Thanks for considering my application.

Sincerely,

Ajay Venkatesh